

Better Executives Work at Better Firms

Identifying Talent Sorting from Mobility Networks

Miklós Koren Ulrich Wohak Krisztina Orbán

6 October 2026, Brown Bag Seminar, CEU

Outline

- 1 Research question and sorting literature
- 2 The measurement problem and existing methods
- 3 Hungarian firms and sparse executive mobility
- 4 Identification from covariance decay
- 5 Estimation with persistent disturbances
- 6 Sorting estimates and revenue implications

Research question

Do better executives work at better firms?

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Object of interest: the correlation

$$\text{Corr}(a_i, z_m). \quad (2)$$

Here $a_i = \log A_i$ and $z_m = \log Z_m$ are latent firm and executive effects.

We combine business outcomes with executive mobility to measure their sorting across the broad corporate sector.

Executive sorting: theory and evidence

Executive assignment models

Perfect positive rank sorting in frictionless competitive benchmarks.

Gabaix and Landier (2008), Terviö (2008).

Evidence from particular organizations

Manager effects and sorting in public-sector offices and retail chains. Estimated sorting can be negative.

Fenizia (2022), Metcalfe, Sollaci and Syverson (2023), Weidmann et al. (2026).

An outcome model with a single executive

Firm i has executive m . Its log real revenue, demeaned by two-digit industry $\times$ calendar year, is

$$y_{im} = a_i + z_m + \varepsilon_{im}. \quad (3)$$

- a_i : persistent firm productivity.
- z_m : persistent executive talent.
- ε_{im} : a disturbance specific to their match.

Object of interest: sorting between the firm and executive components.

AKM and the leave-out correction

AKM (in a nutshell): estimate firm and executive effects

Mobility allows comparisons across firms holding the executive effect fixed. Executive changes allow comparisons within firms holding the firm effect fixed. Combine these comparisons to estimate effects and their covariance.

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Variances and covariances of estimated effects include estimation noise, even when the effects themselves are estimated without bias. With few informative moves, firm and executive effects are estimated less precisely, magnifying this bias.

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KSS (in a nutshell): correct the bias by leaving observations out

KSS corrects this bias using leave-out predictions. The relevant comparisons must remain connected after the observation is removed.

Abowd, Kramarz and Margolis (1999); Kline, Saggio and Sølvesten (2020).

Hungarian firms and executive careers

Administrative records, 1990–2022

Corporate-court records of executive appointments.

Annual firm accounts, including revenue and employment.

1,036,226 firms

1,304,563 executives

1,937,520 unique links

Coverage and executive histories

Data coverage: the universe of Hungarian corporations.

Executive population: chief executives identified among registered company representatives.

Linked careers: follow the same executive across firms and track leadership changes within firms.

We observe who runs each firm, when they serve, and how the firm performs.

Linked corporate-court and accounting records. Counts before graph restriction.

Sparse mobility and leave-out sample restrictions

Redundant mobility is rare

Connected components	514,084
Firms in largest component	33.0%
Executives in largest component	29.6%
Links that are bridges	81.8%

A bridge is the only link connecting two parts of the network. Removing it disconnects them.

Left: current linked sample, before graph restriction. Right: historical match-level leave-out audit; counts are unique firm-executive pairs.

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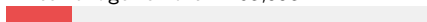
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Match-level leave-out sample

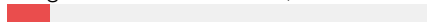
Finite-outcome pairs: 1,732,154



After bridge removal: 265,995



Largest leave-out block: 175,661

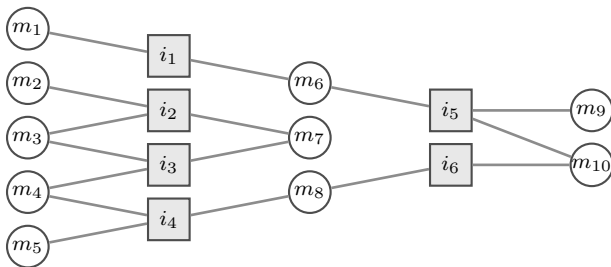


Largest retained block: 10.1% of pairs.

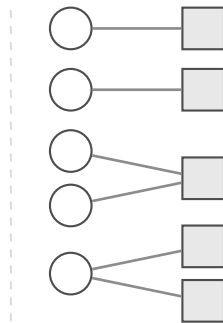
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The executive–firm mobility graph

- Firms (squares) and executives (circles) are nodes.
- An edge is an observed firm–executive link.
- Observed revenue outcomes live on edges; latent effects live on nodes.
- Mobility creates paths between firms and executives that never directly meet.



Schematic mobility network



Many small components

Our approach: the joint distribution of latent effects

- We model the joint distribution of latent firm productivity and executive talent.
- We do not first estimate one fixed effect for every firm and executive and then debias their covariance.
- Stack firm and executive effects:

$$x = (a_1, \dots, a_{N_f}, z_1, \dots, z_{N_m})^\top. \quad (4)$$

- Conditional on the observed mobility network G , assume

$$x \mid G \sim \mathcal{N}(0, Q^{-1}). \quad (5)$$

- The precision matrix Q is sparse and follows the observed firm–executive network.
- This is a **Gaussian Markov random field**.

What does the Markov restriction mean?

- **Random field:** a latent random variable is attached to every node.
- **Gaussian:** these latent variables are jointly normally distributed.
- **Markov:** conditional on a node's direct neighbors, the remaining nodes provide no additional information about its latent effect.

A simple path in a tree



Linked firm-executive correlations:

$$\begin{aligned}\text{Corr}(a_{i_1}, z_{m_1}) &= \text{Corr}(z_{m_1}, a_{i_2}) \\ &= \rho.\end{aligned}\quad (6)$$

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knowing firm productivity a_{i_1} would provide no additional information about a_{i_2} :

$$a_{i_1} \perp a_{i_2} \mid z_{m_1}. \quad (7)$$

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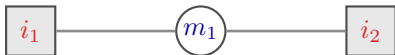
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The firms can be correlated when talent is unobserved. On this Gaussian Markov tree,

$$\text{Corr}(a_{i_1}, a_{i_2}) = \rho^2. \quad (8)$$

Two connected outcomes: identification intuition

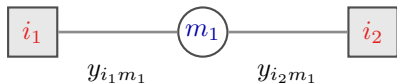
A stylized tree: one executive serves two firms. Disturbances are independent of latent qualities and one another.



Circle: executive m . Squares: firms i .

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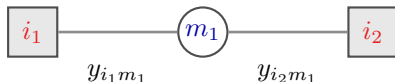
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$$\begin{aligned} y_{i_1 m_1} &= a_{i_1} + z_{m_1} + \varepsilon_{i_1 m_1}, \\ y_{i_2 m_1} &= a_{i_2} + z_{m_1} + \varepsilon_{i_2 m_1}. \end{aligned} \quad (9)$$

Tree benchmark: linked correlation ρ and common type variances σ_a^2, σ_z^2 .

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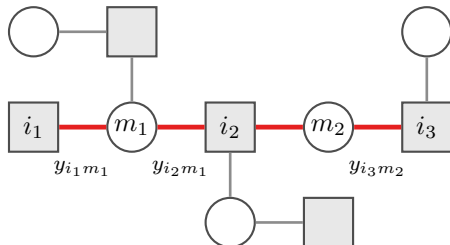
Tree benchmark: linked correlation ρ and common type variances σ_a^2, σ_z^2 .

Expand the outcome covariance

$$\begin{aligned} \text{Cov}(y_{i_1 m_1}, y_{i_2 m_1}) &= \text{Cov}(a_{i_1}, a_{i_2}) + \text{Var}(z_{m_1}) \\ &\quad + \text{Cov}(a_{i_1}, z_{m_1}) + \text{Cov}(a_{i_2}, z_{m_1}) \quad (10) \\ &= \rho^2 \sigma_a^2 + \sigma_z^2 + 2\rho\sigma_a\sigma_z \\ &= (\rho\sigma_a + \sigma_z)^2. \end{aligned}$$

One covariance alone does not identify sorting.

Covariance decay on a tree



One path connects each pair of nodes

C_2, C_4 : firm-to-firm distances of 2 and 4 links

On a tree, each additional link transmits a factor ρ .

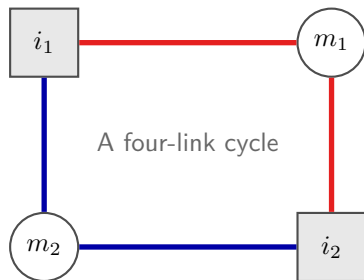
$$\begin{aligned} C_2 &= \text{Cov}(y_{i_1 m_1}, y_{i_2 m_1}) \\ &= (\rho\sigma_a + \sigma_z)^2, \end{aligned} \quad (11)$$

$$\begin{aligned} C_4 &= \text{Cov}(y_{i_1 m_1}, y_{i_3 m_2}) \\ &= \rho^2(\rho\sigma_a + \sigma_z)^2. \end{aligned}$$

$$\boxed{\frac{C_4}{C_2} = \rho^2} \quad (12)$$

Comparing covariances removes the scale of heterogeneity.

Cycles create additional paths



Two routes connect the same firms

$$i_1 \longrightarrow m_1 \longrightarrow i_2$$

$$i_1 \longrightarrow m_2 \longrightarrow i_2$$

Conditioning on one executive no longer separates the two firms.

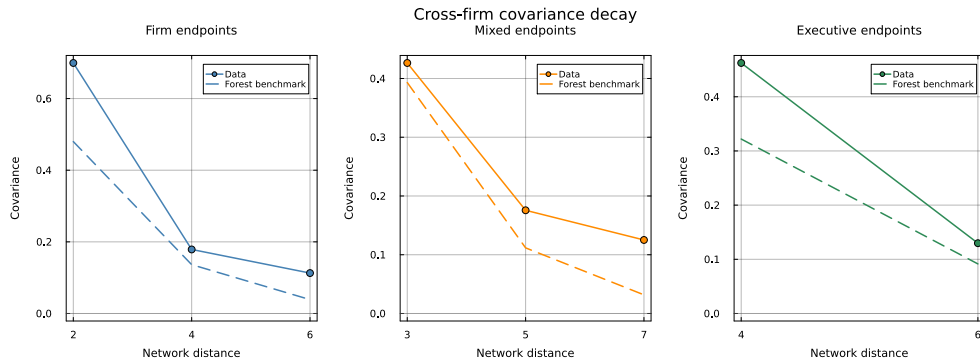
Distance alone no longer determines covariance.

The GMRF uses Q^{-1} to account for all paths in the retained graph.

Before fitting, we remove selected cycle-closing links to obtain a valid covariance matrix.

Stylized cycle. On cyclic graphs, linked marginal correlations are computed from Q^{-1} .

Covariance decay in the data



Markers: industry $\times$ year adjusted outcome covariances. Dashed curves: forest benchmark at the preferred AR(1) parameters.

Descriptive path sample. Single-executive endpoint matches in different firms; linear vertical scale.

Maximum likelihood estimation

The outcome model implies a joint Gaussian distribution for demeaned log revenues:

$$y = Vx + \varepsilon, \quad \text{Var}(y \mid G) = VQ^{-1}V^{\top} + E. \quad (13)$$

- V maps each outcome to its firm and executive effects.
- E is the disturbance covariance matrix; initially, $E = \sigma_{\varepsilon}^2 I$.

We estimate $(\sigma_a, \sigma_z, \rho, \sigma_{\varepsilon})$ by maximum likelihood.

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A shameless plug: our Julia package `BipartiteGMRF.jl`.

Persistent firm-level disturbances

Successive executive matches at the same firm may share persistent business shocks.

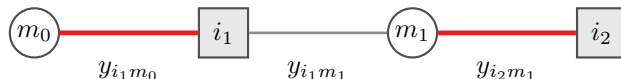
Their covariance combines persistent firm quality with **serially correlated disturbances**.

Our AR(1) specification orders successive executive matches within each firm:

$$\text{Cov}(\varepsilon_g, \varepsilon_h) = \sigma_\varepsilon^2 \eta^{|r_g - r_h|} \quad \text{within a firm.} \quad (14)$$

The gap counts observed matches, not calendar years. Disturbances are uncorrelated across firms.

Identification with correlated disturbances



- $y_{i_1 m_0}, y_{i_1 m_1}$: same firm. Their covariance includes persistent disturbances.
- $y_{i_1 m_0}, y_{i_2 m_1}$: different firms and executives. No common residual shock **under cross-firm error independence**.
- $y_{i_1 m_1}, y_{i_2 m_1}$: different firms, same executive. The introductory example also avoids within-firm error covariance.

Cross-firm covariances retain information about sorting despite within-firm persistence.

Sorting estimates across specifications

Method	Local parameter ρ	$\text{Corr}(a_i, z_m)$
Independent shocks	0.372	0.374
Firm averages	0.562	0.572
AR(1) shocks	0.533	0.541

The preferred AR(1) model estimates a firm–executive correlation of **0.541**.

The preferred AR(1) fit estimates adjacent-match shock correlation $\hat{\eta} = 0.495$.

Industry $\times$ year adjusted outcomes. Same retained graph; firm averages use a different outcome aggregation.

Sources of revenue dispersion

Component	Variance contribution	Share
Firm productivity	0.958	23.2%
Executive talent	0.029	0.7%
Twice sorting covariance	0.187	4.5%
Match disturbances	2.951	71.6%
Total	4.124	100.0%

The executive contribution reflects variation in executive talent. Sorting contributes twice its covariance with firm productivity.

The disturbance term is the marginal variance of the match disturbance, including in the AR(1) model.

Preferred industry $\times$ year AR(1) fit. Equal weight per observed match; average conditional model variances.

Revenue implications: a lognormal benchmark

Hold fixed the marginal distributions of firm productivity and executive talent, match shocks, and the number of matches.

Use the average moments from the preceding slide in a log-normal accounting benchmark. Set the sorting covariance to zero while holding the marginal variances fixed.

$$\frac{Y_{\text{no sorting}}^{\text{benchmark}}}{Y_{\text{observed}}^{\text{benchmark}}} = \exp\left(-\frac{V_{\text{cross}}}{2}\right) = 0.911. \quad (15)$$

The accounting benchmark implies 8.9% lower revenue.

Holds executive heterogeneity fixed. Uses average moments to represent the observed sector.

Scope of the interpretation

The sorting interpretation rests on the latent dependence model.

The outcome adjustment removes industry $\times$ year means. Geographic factors and remaining common shocks can still generate covariance along mobility paths.

Cross-firm comparisons rely on cross-firm error independence.

Allowing persistent shocks within firms does not accommodate unrestricted common shocks across firms.

Executive moves responding to shocks, ownership decisions, and lasting effects after departure can affect the economic interpretation of the persistent components.

Conclusions

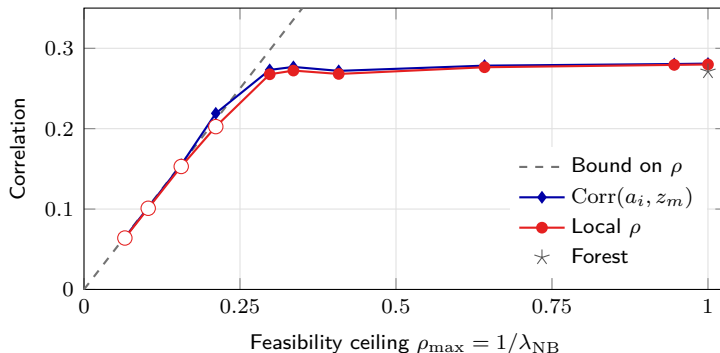
Better executives work at better firms.

Covariance across mobility connections reveals latent sorting in a sparse network.

After industry $\times$ year adjustment, the preferred firm–executive correlation is **0.541**.

Holding firm and executive marginal heterogeneity fixed, the log-normal accounting benchmark implies **8.9%** lower revenue without sorting.

Pruning and sorting estimates



Why prune?

Lower spectral radius widens the admissible range of ρ .

Once the bound is slack

Firm-executive correlations:

0.272–0.281.

Forest benchmark: **0.271.**

Positive sorting remains similar across these pruning thresholds.

Moving right means more pruning. Open circles mark estimates near the bound.

Historical edge-level independent-shock specification. Separate from the preferred industry $\times$ year AR(1) estimates.

Alternative treatments of within-firm dependence

Method	Information used	Within-firm disturbances
Independent-shock likelihood	All match outcomes	Independent benchmark
Cross-firm minimum distance	Selected cross-firm covariances	Unrestricted covariance
Firm-average likelihood	One average outcome per firm	Unrestricted covariance, modeled aggregate variance
AR(1) likelihood	All ordered match outcomes	Common geometric persistence

All maintain cross-firm error independence and the latent dependence model.

An economic foundation for local dependence

Before matches form, firm and executive types are independent in the population.

Complementarity increases the propensity of productive firms and talented executives to match.

Conditioning on observed links creates dependence between their latent types.

A **sparse positive-link approximation**, with quadratic type terms and a valid precision matrix, produces the Gaussian Markov specification.

The approximation omits the complete nonlink denominator. It describes link formation; market clearing would require additional structure.

AKM normal equations and connected comparisons

For the single-executive model, let r index observations of firm $i(r)$ and executive $m(r)$, and let y_r be adjusted log revenue.

$$\hat{z}_m = \frac{1}{n_m} \sum_{r:m(r)=m} (y_r - \hat{a}_{i(r)}), \quad \hat{a}_i = \frac{1}{n_i} \sum_{r:i(r)=i} (y_r - \hat{z}_{m(r)}). \quad (16)$$

- Each effect averages outcomes after subtracting the estimated effect on the other side.
- The equations are coupled: an executive's estimate depends on firms' estimates, which depend on other executives' estimates.
- A normalization fixes the level within each connected component. Disconnected components have separate normalizations.

Repeated observations enter through n_m and n_i . Weighted least squares uses weighted averages.